

LÊ MINH NHẬT

AI Engineer · Applied AI & Interactive Web

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PROFILE

AI Engineer who ships the whole path from model to interface — computer-vision pipelines, LoRA fine-tuning of diffusion models, GPU latency budgets, and the React / Three.js products that expose them. Three months as AI Engineer at STEM Education Technology building explainable mathematics software for grades 1–9, after four months as a Business Analyst turning stakeholder requirements into functional specifications. Trains and gates his own adapters, and ships solo: Android apps on React Native, live web experiences on Cloudflare Workers.

CORE SKILLS

AI & Computer Vision	Python · PyTorch · TensorFlow / Keras · YOLOv8 (detection & pose) · OpenCV · U2Net segmentation · MediaPipe · ByteTrack · face_recognition · NumPy · scikit-learn
Training & Fine-tuning	LoRA / DreamBooth on FLUX.2 Klein (img2img) · Hugging Face diffusers · temporal LoRA (Wan2.1) · BF16 · gradient checkpointing · 8-bit Adam · latent caching · CPU offload · identity-grouped dataset splits · checkpoint acceptance gates · baseline-vs-adapter benchmarking
Generative & Multimodal	FASHN VTON 1.5 · FLUX.2 Klein 4B · Wan2.1 One-to-All Animation · Google Gemini API · pose transfer · image quality gates
MLOps & GPU	CUDA · VRAM budgeting · OOM pilot runs · model warm-up and caching · GPU arbitration · cold/warm P50–P90 latency profiling · Hugging Face datasets · reproducible training logs
AI-Assisted Engineering	Claude Code · OpenAI Codex · MCP servers · authored Claude Code Skills · subagents · agent workflows · spec-driven prompting · AI code review
Web	React 19 · TypeScript · JavaScript · Three.js / React Three Fiber · WebGL 2 · GSAP · Tailwind CSS · Vite · Web Audio · responsive and reduced-motion design
Mobile	React Native · Expo · Java
Backend & Data	Node.js · Express · Flask · FastAPI · Pydantic · SymPy · MongoDB · PostgreSQL · Redis · IndexedDB · REST API
Tooling & Platforms	Playwright · regression gates · Git · GitHub Actions · Linux (Ubuntu) · Cloudflare Workers · Figma · Power BI

AI ENGINEERING TOOLING

Agentic coding tools are daily practice, not an add-on — they are how I deliver multi-stack products at team scope on my own.

- **Claude Code (Anthropic)** — run agentic sessions end to end: **CLAUDE.md** project memory, custom skills and slash commands, subagents for parallel exploration and code review, MCP servers connecting the agent to project tooling, and plan mode before large refactors.
- **OpenAI Codex** — delegate scoped, well-specified tasks to a sandboxed agent; used for second-opinion review, test generation and repetitive migration work across a codebase.
- **Authored five production Claude Code Skills** — packaged my own ML engineering discipline into reusable agent workflows, each enforcing what may and may not be claimed about a model:
 - **japano-fit-trainer** — end-to-end LoRA fine-tuning workflow for FLUX.2 Klein: dataset provenance and licensing, credential hygiene, VRAM-budgeted pilot runs, resumable training, and a hard rule that “fine-tuned” may only be claimed with a licensed dataset, a training log, a checkpoint, validation images from it and a baseline benchmark — otherwise the status stays `MODEL_NOT_TRAINED`.
 - **japano-motion-trainer** — temporal LoRA for Wan2.1 One-to-All Animation: conditioning contract, per-frame identity and garment-consistency gates, joint identity/garment/source-video dataset splits and video licensing rules.
 - **japano-body-measurement** — calibrated anthropometric regression from a single image, returning evidence-bearing confidence intervals instead of false precision, with out-of-distribution rejection and named regression cases.
 - **japano-gpu-performance** — cold and warm P50/P90 latency budgets across every pipeline stage, VRAM ceilings, model warm-up, caching and GPU arbitration, with a rule that inference-only time is never reported as total latency.
 - **img2threejs** — image-to-procedural-Three.js reconstruction through a staged sculpt pipeline and an AI-vision self-correction loop; published Apache-2.0.
- **Working method** — specification first, then generation, then verification. Every generated change is read, run and tested before it lands; AI output is treated as a draft to challenge, never a result to trust.

PROFESSIONAL EXPERIENCE

AI Engineer — STEM Education Technology Co., Ltd.

May 2026 – Aug 2026

- Built Math Vision Lab, an explainable learning engine for grade 1–9 mathematics that turns a typed or photographed problem into a semantic model, a step-by-step pedagogy plan and a synchronised visual scene — so learners see the quantities, relationships and transformations, not only the final answer.
- Designed the deterministic scene pipeline — semantic analysis → decision engine → pedagogy plan → scene → React renderer — covering bar models, fractions, algebra tiles, graphs, Oxyz and 3D geometry, with one shared playback state keeping every representation on the same teaching step.
- Added regression gates and an isolated demo workflow so a change to one visual family cannot silently break another, and unsupported input explains its limit instead of rendering a diagram that contradicts the problem.
- Shipped the EduVision Smart Classroom product front-end: a cinematic Three.js experience — classroom, characters, camera work, digital-twin view and effects — constructed entirely in code with no external 3D asset downloads, deployed on Cloudflare Workers Static Assets and verified across desktop, tablet and mobile.
- Built interactive digital Chemistry Lab experiences for science learning and practice.

Stack React · Three.js · Python · FastAPI · Pydantic · SymPy · PostgreSQL · Redis · Cloudflare Workers

Math Vision Lab demo youtu.be/Ee5RhMHv5AE · EduVision source github.com/minhnhhatdepzai/smartroom

Business Analyst Intern — Mesh, Smart Workplace

Mar 2023 – Jul 2023

- Gathered and analysed requirements from clients and stakeholders, then documented business workflows and functional specifications used directly by the development team.
- Built reporting and data-visualisation dashboards in Power BI, and kept requirements unambiguous between the business and engineering sides.

SELECTED PROJECTS

JAPANO — AI Fashion Commerce Ecosystem

2026

Graduation project · 3 contributors · Android app, storefront, operations console, commerce backend and local GPU services

AI / ML ENGINEERING

- Owned the virtual try-on pipeline: FASHN VTON 1.5 inference with FLUX.2 Klein 4B pose transfer, YOLOv8n-pose and U2Net pre-analysis, and a quality gate that withholds the result when identity, body structure or garment category drifts.
- Fine-tuned a rank-8 / alpha-8 LoRA adapter on FLUX.2 Klein 4B (DreamBooth img2img via diffusers) for fit refinement inside a 16 GB VRAM budget — BF16, gradient checkpointing, cached latents, CPU offload, 8-bit Adam, gradient accumulation 4, at 512 px.
- Ran a 14–28 sample pilot at 20–50 steps to prove real backward passes, finite loss and a loadable checkpoint before a 400–800 step run at LR 1e-4; VITON-HD split 70/15/15 by identity, test set never used to select a checkpoint.
- Defined the acceptance gate — baseline against adapter on identical identity, condition, caption, seed and resolution, scored on identity preservation, body-shape preservation, garment fidelity, fit-effect correctness, artifact rate and p50/p95 latency. Checkpoint 400 accepted; 500 and 600 rejected for artifacts and left disabled by default.
- Built image-based body estimation returning low/high/point intervals with confidence and supporting evidence rather than false precision, with out-of-distribution rejection and user-entered measurements always winning.

PRODUCT ENGINEERING

- Delivered across an Android app (Expo SDK 51 / React Native 0.74), a React 19 storefront, an operations console and a Node 20 + Express + MongoDB backend sharing a single source of commerce truth, with commerce rules enforced server-side.

Stack React Native · Expo · React 19 · TypeScript · Node.js · Express · MongoDB · PyTorch · diffusers · FASHN VTON 1.5 · FLUX.2 Klein · YOLOv8-pose · U2Net

Showcase japano-golden-ticket-showcase.lnhhat1938.workers.dev · App demo youtu.be/D8jqwvPUfFc · Web module github.com/minhnhhatdepzai/JAPANO_web

SAIGON // 77 — Browser-Native 3D Open World

2026

Solo build · playable desktop demo · original procedural systems, no extracted commercial assets

- Built a WebGL 2 open-world vertical slice inspired by contemporary Ho Chi Minh City where walking, driving, traffic, pedestrians, street interactions and city missions all share one continuous playable world.
- Implemented seven playable activities — racing, construction escape, memory bridge, football, prop hunt and maze combat — linked by a shared settlement loop and local career persistence in IndexedDB.
- Added adaptive graphics that respond to the detected renderer and sustained frame performance, keeping a dense simulation readable inside a browser, and validated builds with Playwright.

Stack React · TypeScript · Three.js · React Three Fiber · WebGL 2 · IndexedDB · Playwright

Play saigon-77-open-world.lnhhat1938.workers.dev

K'HO DIGITAL HERITAGE — Interactive Cultural Experience

2026

Solo build · live on Cloudflare Workers · Identity Award, Bàn địa Khởi sinh competition

- Developed a procedural Langbiang landscape and a first-person knowledge world in React Three Fiber, with collectible Flagcards, a digital passport and a 3D brocade service layer for local-value tourism.
- Built a source-aware K'ho cultural atlas — costume, brocade, cuisine, festivals and language chapters — with step-based presentation controls replacing static slides.

Stack React 19 · TypeScript · Three.js · React Three Fiber · Web Audio · Cloudflare Workers

Live k-ho.lnhhat1938.workers.dev · Source github.com/minhnhhatdepzai/K-ho

TASK1 COACH — Local-First English Learning Workspace

2026

Solo build · live · vocabulary, grammar and all four IELTS skills in one workspace

- Built a responsive English practice workspace covering 30 Reading lessons, 30 Listening lessons, 30 Speaking topics and 31 Writing prompts, replacing the scattered documents, audio players and note apps learners normally juggle.
- Kept every draft, answer, playback position, recording and preference on the learner's own device in IndexedDB — no account required — with JSON backup and restore for portability.
- Covered responsive layout, real audio playback, persistence, reduced motion and keyboard-friendly interaction with Playwright regression tests run against the production build.

Stack React 19 · TypeScript · Expo · IndexedDB · Playwright · GitHub Pages

Live minhnhatdepzai.github.io/task1-coach-pages · Source github.com/minhnhatdepzai/task1-coach-pages

PICKO247 — Pickleball Tournament & Fan-Play Hub

2026

Solo product architecture and front-end · live demo

- Implemented the tournament engine: teams and players, group standings recomputed from match results rather than hard-coded, and knockout brackets where winners advance through explicit next-match relationships.
- Integrated Rally 247, a Three.js rally game with ball ballistics, an AI opponent and mobile controls, opening as an overlay so the spectator journey is never cut.

Stack React 18 · TypeScript · Three.js · Web Audio · Tailwind CSS · Cloudflare Workers

Live pickleball247.senaafptpolytechnic.workers.dev · Source github.com/minhnhatdepzai/Pickleball247

HAPPYTOPLAY — Inclusive Social Game Hub

2026

Game Jam entry, FPT Polytechnic · First Prize, Poly Web Game Championship

- Built a mobile-shaped social space with voice rooms, messaging, profiles, rewards and leaderboards, alongside 2D, 3D, microphone- and hand-controlled mini-games.
- Added sign-phrase and speech tools on the Web Speech API and MediaPipe, with local-first fallbacks so the app stays fully explorable when services are offline.

Stack React 19 · TypeScript · Three.js · Motion · Web Speech API · MediaPipe · Flask

Source github.com/minhnhatdepzai/GameJamFPTPolytechnic

PORTFOLIO — Bilingual Interactive Site

2026

Solo build · live on Cloudflare Workers · English / Vietnamese

- Built a bilingual portfolio around a lazy-loaded procedural 3D garden — chameleon, roaming gecko, butterflies, birds and flowers — all code-native, with no remote model or texture downloads.
- Made performance and accessibility explicit: WebGL suspends offscreen and in hidden tabs, mobile and WebGL-unavailable sessions fall back to an animated SVG/CSS habitat, and audio never autoplays.
- Shipped nine deep-linked case-study dialogs and a Cloudflare Workers Static Assets pipeline with a parallel GitHub Pages build.

Stack React 19 · TypeScript · Three.js · GSAP · Tailwind CSS 4 · Vite 6 · Cloudflare Workers

Live minhnhat-portfolio.lnhat1938.workers.dev

VEHICLE COUNTING — Detection, Tracking & Line-Zone Counting

2026

- Detects vehicles and pedestrians with YOLOv8n, tracks them across frames with ByteTrack, and counts each direction across a line zone, writing an annotated output video.

Stack Python · YOLOv8 (Ultralytics) · Supervision / ByteTrack · OpenCV · NumPy · Source github.com/minhnhatdepzai/Count_vehicles

EDUCATION

FPT Polytechnic College — Mobile Application Development

2024 – 2026

University of Information Technology — Computer Science

2020 – 2023

AWARDS & RECOGNITION

Winner	First Prize, Poly Web Game Championship · Identity Award, Bản địa Khởi sinh Competition
Placed	Second Prize, FPT Polytechnic Language School · Third Prize, Poly Coder Club Chatbot Design · Top 5, FPT Polytechnic UI/UX · Top 8, Poly Coder Web Design Hackathon · Top 8, Viet My College Vibe Coding · Semifinalist, EVC Competition (UEF)

KNOWLEDGE SHARING

- **AI-Assisted Development tutoring session** — self-organised, researched and written independently: using AI for ideation and prototyping, then reviewing and challenging the output. [Post](#)
- **Information Security seminar — Broken Authentication** — self-researched and delivered: the phishing attack lifecycle, spoofed-website recognition and defences, demonstrated in a controlled lab. [Post](#)